

RAG System Performance Report

Author: AI Lab

Abstract

This report evaluates the RAG architecture for document QA.

1. Introduction

RAG systems combine retrieval with generation.

Table 1: Experiment Config

Parameter	Value
Model	bge-large-zh
Dim	1024

2. Method

Metrics: HR, MRR, NDCG.

Formula 1: Hit Rate

$HR@k = (1/N) * \sum(1 \text{ if relevant in top-}k \text{ else } 0)$

2.1 Experiment Setup

5-fold cross validation.

3. Results

Hybrid retrieval outperforms pure vector search.

Table 2: Performance Comparison

Method	HR@5
Vector	0.723
Hybrid	0.856

4. Conclusion

RAG is effective. Next: multimodal.